

PYTHON - SE - BACKEND

SOFTWARE ENGINEERING – Introduction to Programming

MODULE - 2

SESSION / ASSIGNMENT – 5

TASKS :

1. Create a simple IPL Fan Bot that takes your favorite IPL team name as input and uses if-else-if statements to print a unique cheer message for each team (e.g., 'Go Mumbai Indians!', 'Chennai Super Kings for the win!'). If the team is not recognized, print 'Team not found!'



```
#include <stdio.h>
#include <string.h>

int main()
{
    char team[30];

    printf("Enter your favorite IPL team: ");
    scanf("%s", team);

    if (strcmp(team, "MI") == 0)
        printf("Go Mumbai Indians!\n");
    else if (strcmp(team, "CSK") == 0)
        printf("Chennai Super Kings for the win!\n");
    else if (strcmp(team, "RCB") == 0)
        printf("Play Bold, RCB!\n");
    else if (strcmp(team, "KKR") == 0)
        printf("Korbo Lorbo Jeetbo!\n");
    else if (strcmp(team, "GT") == 0)
        printf("Go Gujarat Titans!\n");
    else
        printf("Team not found!\n");

    return 0;
}
```

OUTPUT:

When entering correct team name

```
Enter your favorite IPL team: RCB  
Play Bold, RCB!
```

When entering other team name not found on the list of teams

```
Enter your favorite IPL team: GSK  
Team not found!
```

2. Build a Zomato-style food suggestion tool: take the user's preferred meal time ('breakfast', 'lunch', 'dinner', or 'snack') and use a switch-case statement to suggest a popular dish for that time. If the input doesn't match any meal, suggest 'Try some fruits!'.



```
#include <stdio.h>

int main()
{
    int choice;

    printf("\n1. Breakfast\n");
    printf("2. Lunch\n");
    printf("3. Dinner\n");
    printf("4. Snack\n");

    printf("Enter your choice (1-4): ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            printf("Suggestion: Idli and Dosa\n");
            break;

        case 2:
            printf("Suggestion: Veg Thali\n");
            break;

        case 3:
            printf("Suggestion: Biryani\n");
            break;

        case 4:
            printf("Suggestion: Sandwich\n");
            break;

        default:
```

```

        printf("Try some fruits!\n");
    }

    return 0;
}

```

OUTPUT:

1. Breakfast 2. Lunch 3. Dinner 4. Snack Enter your choice (1-4):	Enter your choice (1-4): 1 Suggestion: Idli and Dosa	Enter your choice (1-4): 2 Suggestion: Veg Thali
Enter your choice (1-4): 3 Suggestion: Biryani	Enter your choice (1-4): 4 Suggestion: Sandwich	Enter your choice (1-4): 5 Try some fruits!

3. Create a Flipkart discount calculator that asks the user for the total cart amount. Use nested if statements to check: if amount > 2000, apply 20% discount; else if amount > 1000, apply 10% discount; else, no discount. Print the final amount to pay.

Hint: Use nested ifs to check each discount slab.



```

#include <stdio.h>

int main()
{
    float amount, finalAmount;

    printf("Enter total cart amount: ");
    scanf("%f", &amount);

    if (amount > 1000)
    {
        if (amount > 2000)
        {
            finalAmount = amount - (amount * 20 / 100);
        }
        else
        {
            finalAmount = amount - (amount * 10 / 100);
        }
    }
    else
    {
        finalAmount = amount;
    }
}

```

```
printf("Final Amount to Pay = %.2f\n", finalAmount);

return 0;
}
```

OUTPUT:

if amount > 2000, 20% discount

```
Enter total cart amount: 2100
Final Amount to Pay = 1680.00
```

if amount > 1000, 10% discount

```
Enter total cart amount: 1500
Final Amount to Pay = 1350.00
```

else, no discount

```
Enter total cart amount: 799
Final Amount to Pay = 799.00
```

4. Write a program that takes a user's age and checks eligibility for three things using if-else statements: if age $\geq$ 18, print 'Eligible for Driving License'; if age $\geq$ 21, also print 'Eligible for Credit Card'; if age $\geq$ 25, also print 'Eligible for Car Rental'. Print all applicable messages for the given age.



```
#include <stdio.h>

int main()
{
    int age;

    printf("Enter your age: ");
    scanf("%d", &age);

    if (age >= 18)
    {
        printf("Eligible for Driving License\n");

        if (age >= 21)
        {
            printf("Eligible for Credit Card\n");

            if (age >= 25)
            {
                printf("Eligible for Car Rental\n");
            }
        }
    }
}
```

```
    }  
    }  
}  
else  
{  
    printf("Not eligible for Driving License\n");  
}  
  
return 0;  
}
```

OUTPUT:

if age >= 18, 'Eligible for Driving License'

```
Enter your age: 18  
Eligible for Driving License
```

if age >= 21, also print 'Eligible for Credit Card'

```
Enter your age: 23  
Eligible for Driving License  
Eligible for Credit Card
```

if age >= 25, also print 'Eligible for Car Rental'

```
Enter your age: 27  
Eligible for Driving License  
Eligible for Credit Card  
Eligible for Car Rental
```